The current service model represents a system with a single queue and a single operator. Such a configuration creates a single point of failure (SPOF) and a structural bottleneck that limits the system's throughput. The main issues are high average service latency, inability to process requests in parallel, complete dependence on a single resource. In addition, different banking operations vary significantly in complexity and processing time. Loan processing may take 20-30 minutes, whereas a money transfer typically requires 3-5 minutes. Combining these operations within a single queue results in inefficient resource utilization.

The objective of the system redesign is to increase throughput, defines as the number of customers served per unit of time, while maintaining consistency and reliability. A banking system requires data consistency. The customer's account balance must always reflect the current and accurate state. Therefore, the architecture is designed to prioritize Consistency (CP-oriented) while maintaining an acceptable level of availability.

Baseline: one operator serves 6 customers per hour (average service time is 10 minutes).

Target: five operators working in parallel, resulting in a theoretical capacity of up to 30 customers per hour.

Optimization: Introduction of an express line to reduce average waiting time for simple tasks.

Efficiency: Pre-validation of documents via mobile application to reduce processing time for complex operations.

Result: Expected throughput increase approximately 3-4 times.

Architectural principles:

To achieve this objective, the following architectural principles are applied: horizontal scaling, load balancing, asynchronous processing, fault tolerance, stateless approach.

1)Horizontal scaling-instead of increasing the performance of a single operator, the system introduces multiple operator instances capable of working in parallel. This eliminates the structural bottleneck.

2)Load balancing-customer distribution among operators is performed based on current workload, average processing time of service types. The load balancer ensures reduced waiting time and balanced resource utilization.

3)Fault tolerance-operator redundancy, centralized data storage, real time workload monitoring. These measures eliminate the single point of failure.

4)Stateless approach-operators don't store customers state locally. All customer data is maintained within a centralized banking system. The advantages are any operator can serve any customer and system resilience against individual workstation failures.

Logical System Architecture, layered view:

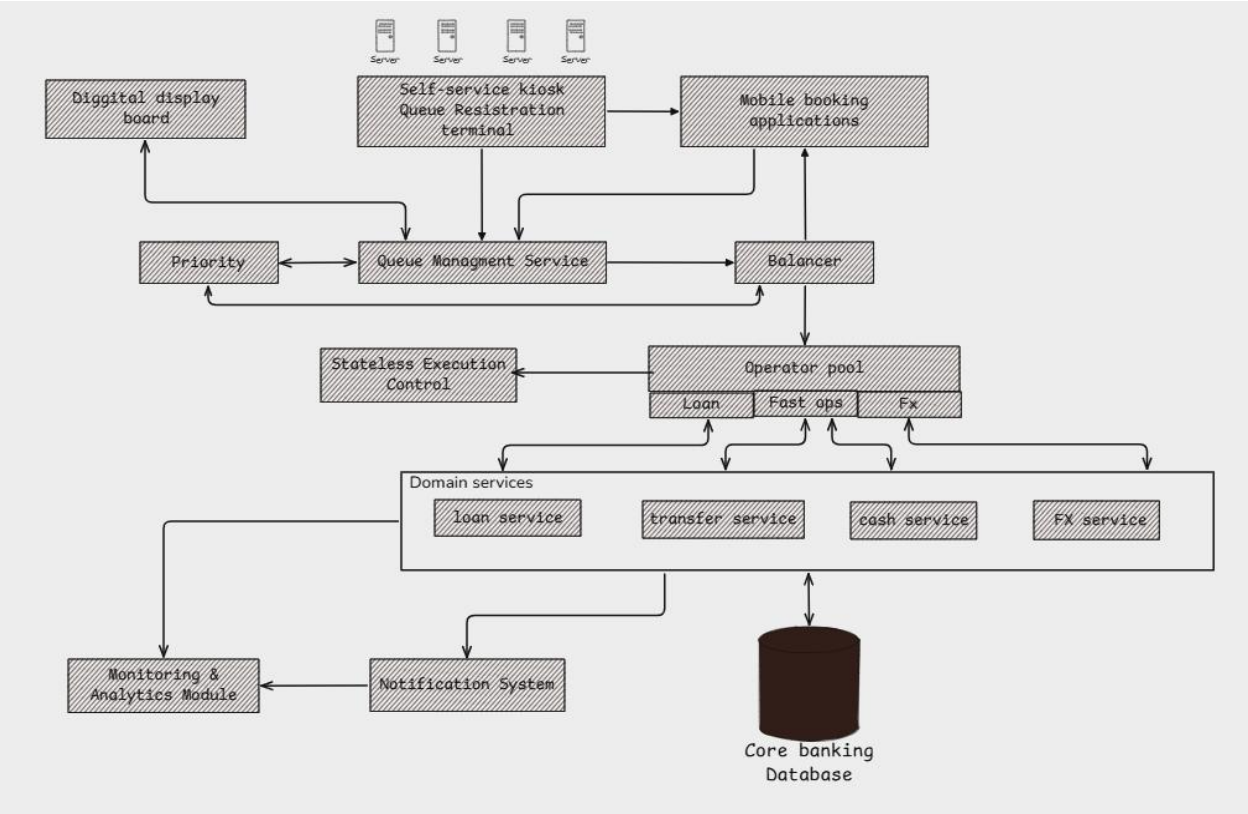
Presentation Layer: self-service kiosks, mobile pre-booking application, digital display boards.

Application Layer: queue management module, request routing engine, prioritization module.

Domain Layer: loan service, money transfer service, cash operations service, currency exchange service.

Infrastructure Layer: core banking database (CP-oriented), notification system (SMS), monitoring and analytics module.

Figure 1. Logical Architecture of the Bank Branch Queue Optimization System



In conclusion, the proposed eliminates the structural bottleneck, enables horizontal scalability, improves fault tolerance, reduces average waiting time, and significantly increases overall branch throughput. The solution is based on distributed system principles and modern approaches to designing scalable and high load service infrastructures.